

SOFTWARE ENGINEERING

Code
Data Structures
Documentation

Characteristics of Software

Software → Set of instructions
→ general instruction used to fulfill the requirements

1) software is engineered not constructed

① instructions

② Data Structure

③ Documentation

Substantionality

→ NO short program

→ software (solution)

→ project management

→ quality management

For big, Authentic domain

Documentation

→ start with idea

→ end with absolution

"Do what is documented and Document what is done."

Software Requirements

Time

transcript

CMS

Complexity

Expertise (people)

Functional req. are the basic req which are known as business logic and normally these are stated by customer SE → S/W

Software works in

Layers

Front end layer

Business logic layer

Database layer

retrieve, return refined

Triple constraints

Calculator → Arithmetic operations

Types

Scope (requirements)

→ should be substantial

Functional Requirements
Business Logic

Non-functional req.

are the limitations or constraints applied on functional reqs.

cost (customer purchase is diff from cost)

Data Structure V/S Database

containers, vessels to hold the data according to the needs

to store the data permanently (persistently)

Scientific req. of Scientific calculator is non-funct req. of calculator.

e.g. : chair in class (no cupboard) created on runtime, used (temporary) and flushed.

Cash

Cash, Card

Cash, Card, online

e.g. : Array, Tree, Stack graph, WSB

Non functional req makes the software

"Security" is not functional req. → MCBs

✓
MTWTFSS

Date: 23-sept

Documentation → Report
→ Maintain $\frac{1}{2}$
→ Document (planning)

Group of people
Substantially
for Remembrance
Synchronized $\frac{1}{2}$
Focused & well-directed work.

→ Technical Documentation

SDLC: Software Development Life Cycle.

Analysis (what)

Design (How) → Logic design $\frac{1}{2}$

code → Logic of code $\frac{1}{2}$

Test → If it works or not

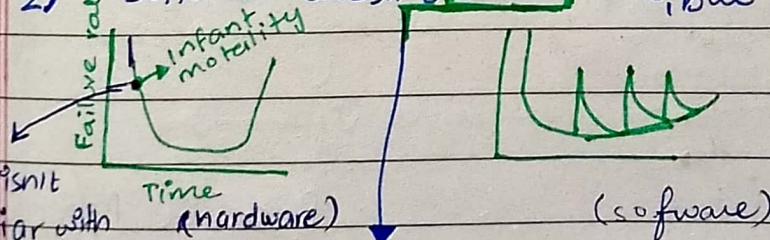
→ User Documentation:

characteristic of software:

1) → software is engineered, not constructed:

→ Its an intangible thing, engineering is used

2) Software doesn't wear out, but it deteriorates.



User isn't familiar with the way of usage of hardware.

Software engineered according to funct and non funct reqs, it will work life time and doesn't wear out.

→ creates:

Changes requested by customer.

constraints are changed. change in requirements. Hardware is changed.

3) Although industry is moving towards Component Based Development, most software is custom built.

4) Software is complex. (not in book)

Software is an abstraction of real life.

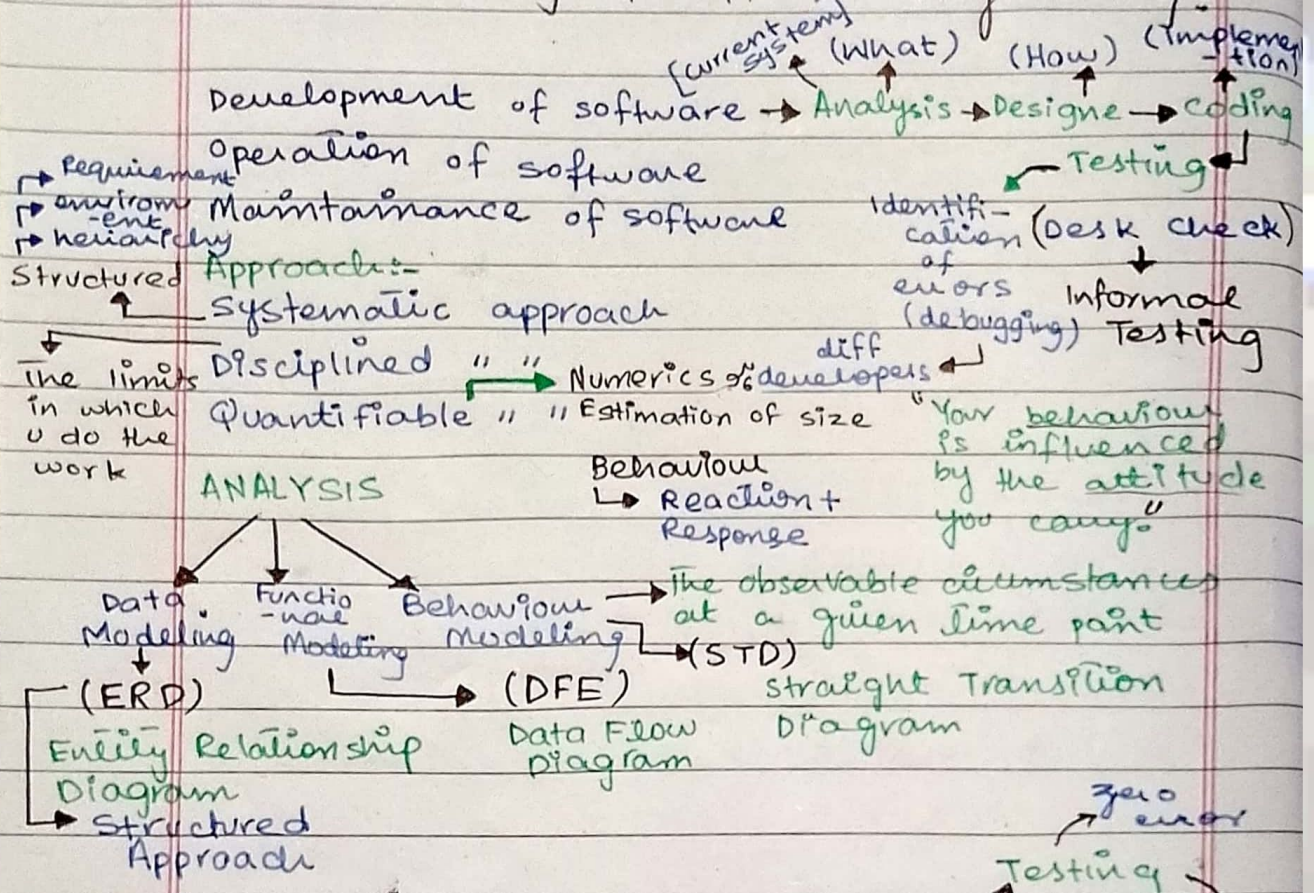
itel S23 4G

"If you're too big to do small things
Then you're too small to do big things"

MTWTFSS

Date: 25-sept

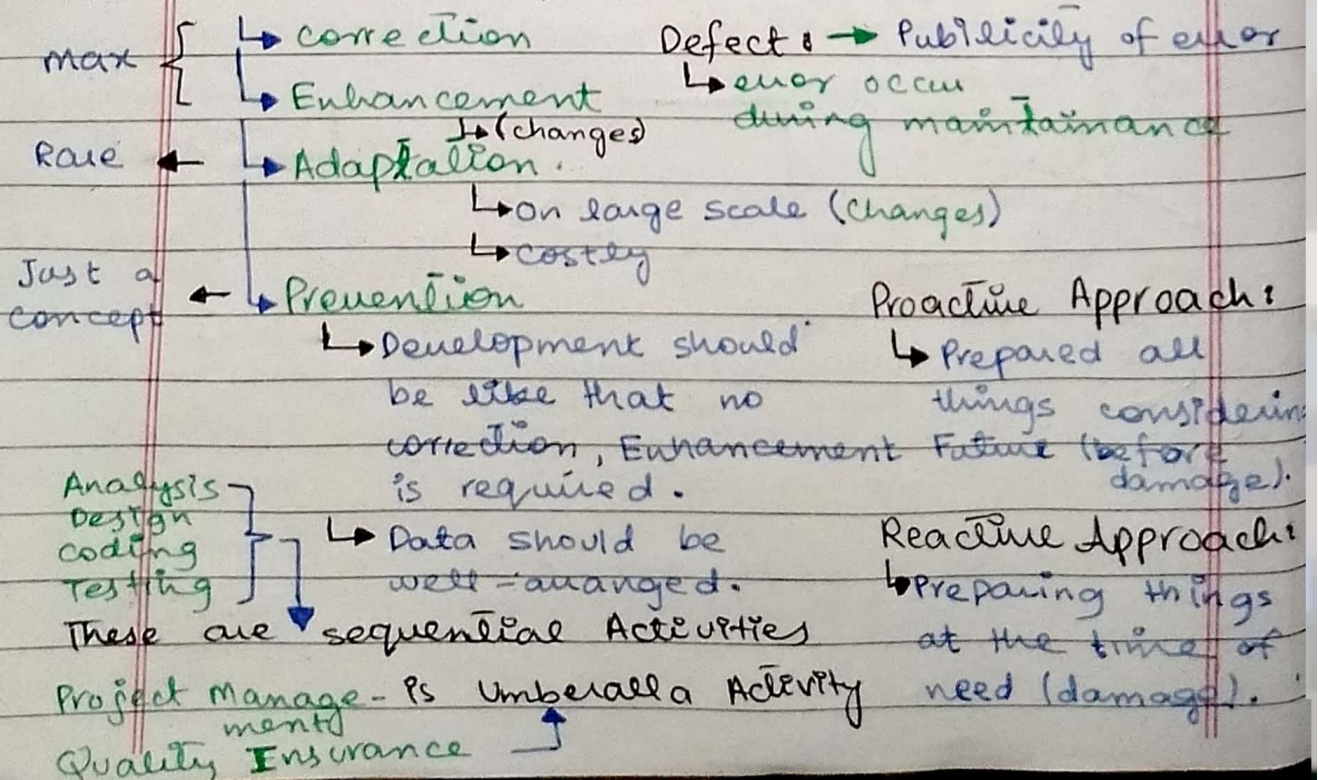
Understanding of Software Engineering:



Operation of software:

↳ User using the software

Maintenance of software:



Software Process And Software Process Models.

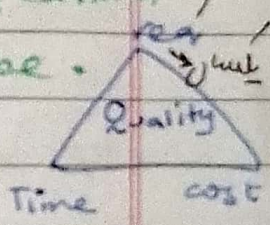
→ **Process** → is the sequence of steps to achieve a particular goal.

→ **Product** → result, Goal Achievement
 ↳ Quality Product/software
 Customer satisfaction
 Quality oriented
 development of software

Software Engineering: A Layered Technology:

↓ **Quality** → definition
 ↳ Customer satisfaction
 ↳ Fitness for use.
 ↳ Continuous improvement
 Quality Focus.
 Each time the excellence of the things cannot be quality. But sometimes, it can be.
 Quality Focus → Bed Rock
 ↳ Glue

Quality is not the absolute concept. (levels)



Process → SDLC (simple)

5 Situations {
 When the req.s are clear → LSM (Linear sequential model)
 when the req.s are not clear → Prototyping
 when the time is short → RAD
 For commercial software (development) → Incremental model
 when there are risks → Spiral model

Method's → How to's
 ↳ suitable here challenge

↳ structured Analysis (Part of Analysis)
 ↳ Data Modeling → ERD
 ↳ Functional Model (DFD) → DFD
 ↳ Behaviour Modeling → STD

E.g.: Flowchart, Diagram, Documents
 Pseudocodes

Tools: → CASE (Computer Aided Software Engineering)

↳ Support of methods
 ↳ MS-Visio
 Application, software usage

MTWTFSS

Simulation
 ↳ complete copy
 ↳ API force simulation
 ↳ game simulation

Date: _____

N

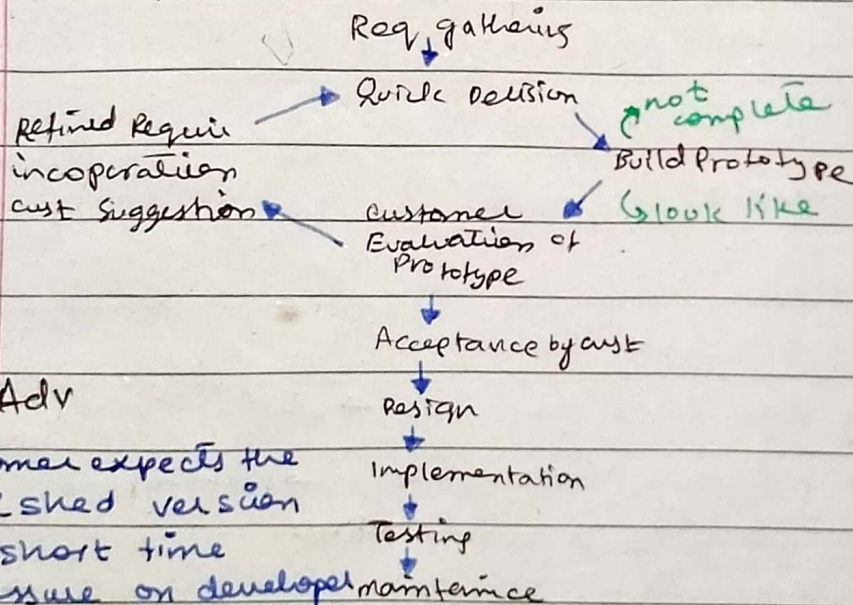
Process V/S Process Model
 ↳ Generic ↳ Particular

LSM → Linear (flow)
 waterfall Model (top-bottom flow)
 classic life cycle (first)
 SPLC (derived from classic)

Foundation of all models

Faults
 Dis-Adv [Not a iterative process model
 Not evolutionary

Prototyping? Not only a model but also a technique to gather reqs.
 complexity, Not cooperative customer
 Sample (show)
 Iterative



Dis-Adv

↳ Customer expects the finished version in short time
 (Pressure on developer to take short cuts at expense and maintainability)

↳ compromises for Speed

↳ In appropriate tools may be used
 The efficient algorithms may become per integral part of system.

MTWTFSS

Diagram making Tools

MS-VISIO
Smart Draw
Rational-ROSE

5 generations of PL.

Assembly
C++
VB
FORTRAN

Date: 2-0ct

7-0ct

RAPID APPLICATION DEVELOPMENT MODEL (RAD)

special case

- Time management on personal level. (PTM)
- Technical Time Management (TTM)

Rapid Fire Activities

- Personal Time management
- Team Player.
- Expert & Experienced

Technical Time Management :-

⇒ Increase the resources (cost will be increase)

⇒ Component-based Development (Parallel work)

↓ contains
(Module) 2 things

independent functionality.

Appropriate Interface

Communication channel through which information is possible

Code (Application-generation) should be used. Programming language

Python
PHP, Java, Dotnet
edition

Prototype

Business modeling

Data modeling

Process modeling

Application generation

Testing & Terminology

Automatic Testing krni ha

Testing &

(turnover)

All components are integrated after testing → after test over again testing then product delivery

Automated tools create dummy user within sec's

exchange of data

Association

Avoid dependency (تلاقی) to reduce complexity